University of St.Gallen

Fundamentals and Methods of Computer Science

Exercise week 2

From insight to impact.

Kursübersicht – Course Overview

Date (L/GE)	Topic (L/GE)	Exercise/E	Coaching/CO	Who	Book Chapter
L01: 19.02.2026 **GE01:** 20.02.2026	Intro, Course Overview, Computing Basics	Setting up Python environments for Assignment 1	*no coaching*	Simon Mayer, Stephan Aier	--
L02: 26.02.2026 **GE02:** 27.02.2026	Programming I	Quiz, Assignment 1: Programming Basics project work is explained	23.02.2026	Stephan Aier	2, 3, 4
L03: 05.03.2026 **GE03:** 06.03.2026	Programming II	Quiz, Assignment 2: Programming I *project groups are defined*	02.03.2026	Stephan Aier	3, 4, 5
L04: 12.03.2026 **GE04:** 13.03.2026	Programming III	Quiz, Assignment 3: Programming II	09.03.2026	Stephan Aier	5, 6, 11
L05: 19.03.2026 **GE05:** 20.03.2026	Programming IV	Quiz, Assignment 4: Programming III *optional group presentation on project idea I*	16.03.2026	Stephan Aier	10
L06: 26.03.2026 **GE06:** 27.03.2026	Distributed Systems	Quiz, Assignment 5: Programming IV *optional group presentation on project idea II*	23.03.2026	Simon Mayer	--
--	*semester break*				
L07: 16.04.2026 **GE07:** 17.04.2026	Databases, Structured Data, SQL	*Quiz, Assignment 6: Distributed Systems* *optional group presentation of MVP of the project I*	13.04.2026	Stephan Aier	17
L08: 23.04.2026 **GE08:** 24.04.2026	Data Science I: Intro	Quiz, Assignment 7: Databases, SQL *optional group presentation of MVP of the project II*	20.04.2026	Stephan Aier	7
L09: 30.04.2026 **GE09:** 01.05.2026	Data Science II: Data Wrangling and Visualization	Quiz, Assignment 8: Data Science I	27.04.2026	Stephan Aier	
L10: 07.05.2026 **GE10:** 08.05.2026	Data Science III: Machine Learning	Quiz, Assignment 9: Data Science II	04.05.2026	Bernhard Bermeitinger	15
Auffahrt **L11:** 15.05.2026	Data Science IV: Machine Learning	Quiz, Assignment 10: Data Science III *final group presentation of project result*	11.05.2026	Bernhard Bermeitinger	15 (16 as an outlook)
L/GE11: 21.05.2026 **Finale:** 21.05.2026	Data Science IV: Machine Learning; Summary and presentation and voting on top-3 projects	Quiz, Assignment 11: Data Science IV	18.05.2026	Bernhard Bermeitinger, *all*	

Elemente, die direkt oder indirekt die Benotung beeinflussen, sind *rot gedruckt.*

Today's Menu

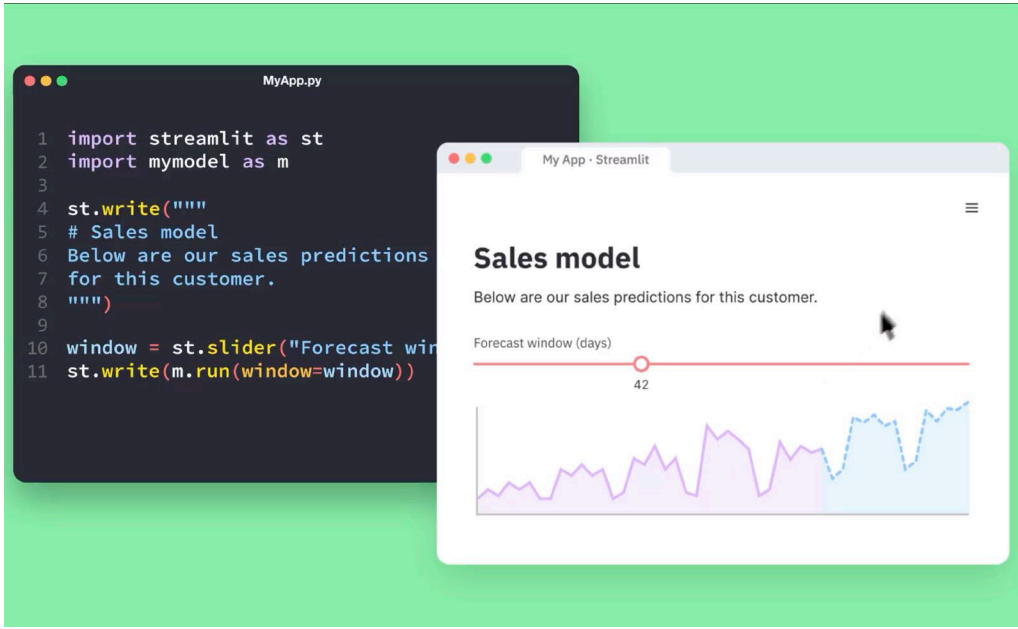
1. Group Project Briefing
2. Discussion of Assignment 1 and the respective Quiz
3. Brief summary of Skills necessary to solve Assignment 2
4. Working on Assignment 2

Group Project - Briefing

Beispiel aus dem letzten Semester

Idea and goals

- You build a **web application** in Python using https://streamlit.io.
- The idea is to apply and **strengthen your skills** developed in this course to **solve a «business» problem**.
- This will very likely require you to **extend your knowledge** provided in lectures and exercises, driven by the particular needs of the selected problem.
- It will also need some **planning**, **structuration**, and **management** of the group work in rather large groups.



Group Project – Overview

Idea

Groups (5 people within the **same exercise group**) build a small web app using **Streamlit**.

Also:
Group formation from 27.02.2026 12:00 till 05.03.2026 23:59 on Canvas
https://learning.unisg.ch/courses/26148/groups

Deliverables

14.05.26 23:59

Max. 4 min. Video
&
Working and well documented source code of the application

Also:
- Presentation of video & 5 min. Q&A (15.05.)
- Top-3 present (21.05.)

Grading

20% of your final grade

Mandatory Requirements (see next slide)

Every requirement (1-8) is evaluated on a 4-point scale, representing points

- 0 – requirement is not met / feature does not exist
- 1 – basic implementation
- 2 – good implementation
- 3 – outstanding implementation, far beyond the level of this course

Requirements and Grading

Requirements relevant for evaluation/grading (mandatory)

1. A problem that is solved by the application is clearly stated (e.g., business or consumer use case)
2. The application uses some data that is loaded via an API and/or provided via a database
3. The application visualizes some data that serves the use case
4. The application allows for some user interaction, e.g., adding additional data, selecting certain data, running certain data analyses
5. The application implements some machine learning
6. The source code is well documented by comments in the source code
7. The contributions of each team member are documented (e.g., contribution matrix).
8. The result is presented and **demoed** in a 4-minute video. The video is not allowed to use AI-generated voice overs.

Optional extensions not relevant for grading but for the "beauty" of the result

- The application is made publicly available to users, e.g., via the streamlit platform
- … anything you ever wanted to implement

Grading

Every requirement (1-8) is evaluated on a 4-point scale, representing points

- 0 – requirement is not met / feature does not exist
- 1 – basic implementation
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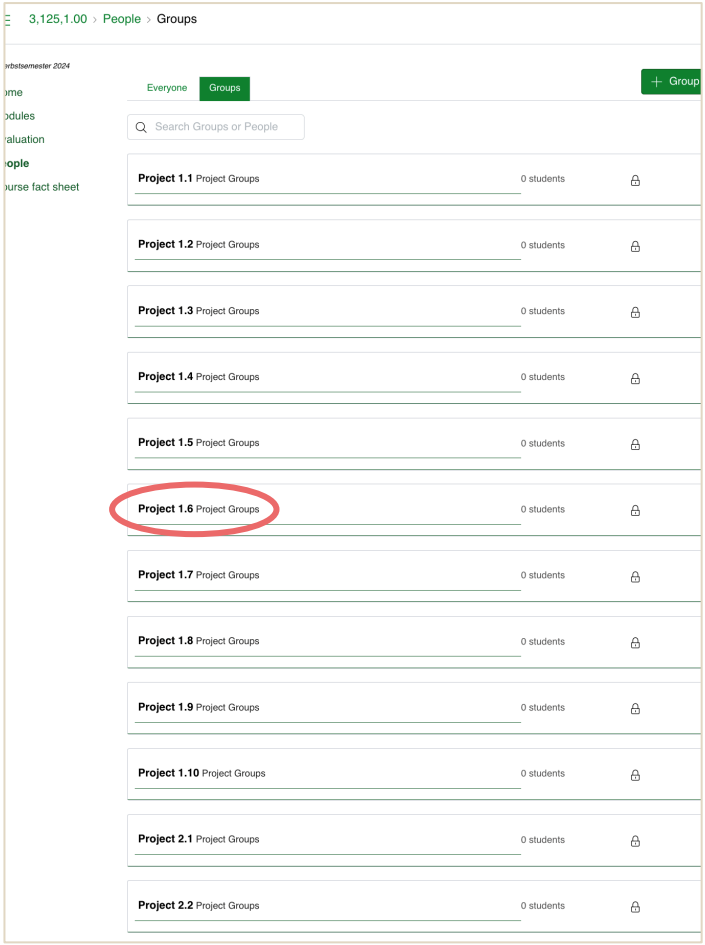
Project evaluation for final grade

0 points	= 0%	9 points	= 56.25%
1 point	= 6.25%	10 points	= 62.5%
2 points	= 12.5%	11 points	= 68.75%
3 points	= 18.75%	12 points	= 75%
4 points	= 25%	13 points	= 81.25%
5 points	= 31.25%	14 points	= 87.5%
6 points	= 37.5%	15 points	= 93.75%
7 points	= 43.75%.	≥16 points	= 100%
8 points	= 50%		

Be careful to reference everything, that is not yours (e.g., other people's contributions, source code, etc.)
See also citation rules for GenAI at HSG: https://universitaetstgallen.sharepoint.com/sites/PruefungenDE/SitePages/en/Working-with-AI.aspx#citation-rules

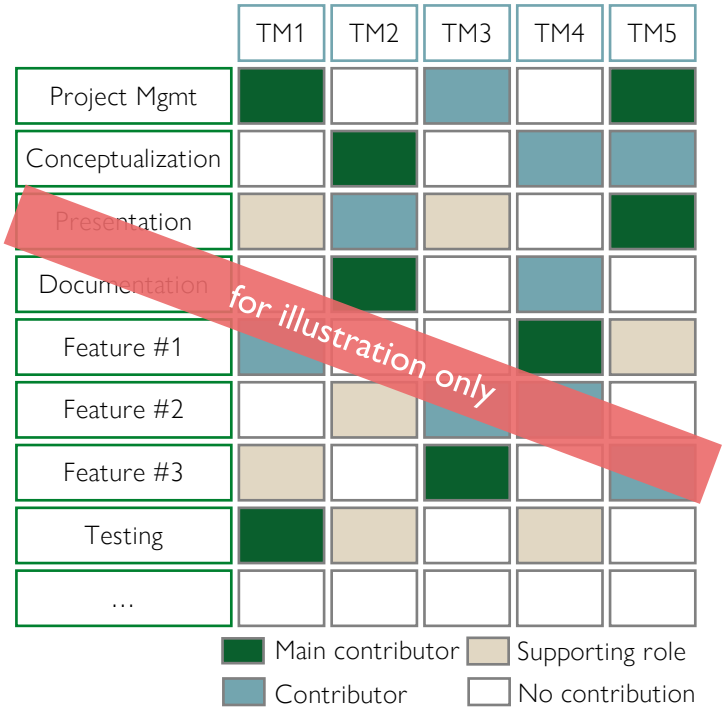
Project group self assignment

- Teams of **max. 5 students**
- **10 student groups** per exercise group
- **Every member** of a student group needs to be in the **same exercise group**
- **You will self assign** to predefined groups on the main canvas room of our course (lecture)
- Groups' naming pattern reflects *exercise group* and *student group* within this exercise group, e.g., **Project 1.6** reflects **exercise group 1** and **student group 6** in that exercise group.
- Please *only* assign yourself to a project of **your official exercise group** – otherwise we need to remove you from that group.



Team structure

- Think about, how to **structure and organize** work in your team.
- What **roles** and **responsibilities** may be reasonable? But remember: writing code is the main idea of the project.
- Given the team size, this will be a major challenge and will require **dealing with conflicts**.
- **Escalate** *unsolvable* conflicts to your tutor **early**. Such a conflict may be unresponsive team members.
- **Document individual contributions** in a contribution matrix.



Collaboration and communication

- Think about **how you want to collaborate** in your team, e.g., whether/when to have
  - Scheduled meetings with every team member (e.g., apps.unisg.ch/gmr)
  - Ad-hoc meetings in subgroups
  - … and what to discuss online, what in presence

- Think about your **collaboration infrastructure**, e.g.,
  - Setting up a team in **MS Teams**
  - Using the group environment in **canvas**
  - Other communication channels
  - (setting up a *github* project, this is really optional, but we have video tutorials for that)

- Build a **respectful** and **inclusive** team environment

Deliverables

1. **4-minute** (max.) **video** on your project idea, process, contributions and result including a short demo of your application.
2. The working and well documented **source code** of your application. Make sure you include all the resources such as pictures, that your program possibly uses. We need to be able to run your application.

Both deliverables need to be uploaded by **Thursday, 14.05.2026, 23:59** midnight to Canvas.

- Each group will show their video in their exercise group on Friday, 15.05.2026 and will have a Q&A session afterwards (*mandatory*).
- The selected top-3 groups will present their solution in the lecture session on Thursday, 21.05.2026 (*mandatory*).

Be careful to reference everything, that is not yours (e.g., other people's contributions, source code, etc.). Be aware of HSG's regulations on the use of GenAI. See exercise session for details.

Milestones: mandatory*/optional**?

Week 03: Group definition / assignment, Q&A for project

Week 04

Week 05: Presentation of project idea / approach part 1** → Feedback (max. 10 minutes/group)***

Week 06: Presentation of project idea / approach part 2** → Feedback (max. 10 minutes/group)***

BREAK

Week 07: Presentation of MVP part 1** → Feedback (max. 10 minutes/group)***

Week 08: Presentation of MVP part 2** → Feedback (max. 10 minutes/group)***

Week 09

Week 10

Week 11: Upload of solution, presentation video (incl. demo), Q&A session/group*

Week 12: Presentation of Top-3 groups (lecture session)

*) You *MUST ATTEND* these exercise session. If you can't make it, you have to discuss this with your group and your exercise session tutor in advance. Not doing so may lead to exclusion from the group.

**) For the first two presentations there is an early (groups 1-5) and a late slot (groups 6-10). You chose during group definition whether you prefer early feedback (groups 1-5) or more time to prepare (groups 6-10).
Those presentations are *OPTIONAL*, if you want to present to your exercise tutor,
(i) you need to register your presentation one week in advance with your tutor AND
(ii) all group members must to be present.

***) The 10-minute slot is for presentation **and** feedback. This is not a lot of time, you decide on how much you will present, i.e., how much time there will be for feedback.

Resources and support

Recources are widely available on the internet, e.g.,

- Streamlit documentation: https://docs.streamlit.io/
- Streamlit cheat sheet: https://docs.streamlit.io/library/cheatsheet
- Streamlit tutorials such as: https://30days-tmp.streamlit.app/
- Inspirational examples as in the Streamlit gallery: https://streamlit.io/gallery
- Examples from lectures/exercises
- Optional tutorial videos on using *github* (available via Canvas)
- … more general questions, e.g., on Stackoverflow, ChatGPT, …

Support is available in many flavors, e.g.,

- Your tutors in the exercise sessions
- Tutors in the coaching sessions
- Professors in and around lectures and guided exercises

Plagiarism in Coding – Acceptable Practices

Copying or adapting code with **proper attribution** is acceptable. Here is what that looks like:

Docs, Google, Stack Overflow

- Learning about a specific function – no citation needed
- Using small snippets to learn a method – no citation needed
- Using larger snippets for a specific task **– cite the source**

Friends, Tutors, Fiverr

- Asking questions to understand concepts – no citation needed

GenAI Models (ChatGPT, Claude & co.)

- Asking questions to learn – no citation needed
- Having it write larger code for you **– cite the source**

Plagiarism in Coding – Unacceptable Practices

The following practices are **not acceptable** and may constitute plagiarism:

- Using code snippets from any source **without proper citation**
- Having someone else (friend, tutor, Fiverr, etc.) **write code for you**
- Having AI (ChatGPT & co.) **generate code without citing it**
- Presenting **someone else's work as your own**, even with modifications

Always follow the University of St. Gallen guidelines on plagiarism and academic integrity.

A Note on ChatGPT (& co.)

If you let ChatGPT code your whole program & ***cite this appropriately***, this is …

- **fine** from a plagiarism point of view
- **bad** from a contributional point of view

In programming you as a programmer …

- **don't need** to reinvent the wheel
- **need** to put the code together in a new way, such that it solves your specific problem

If your solution lacks contribution, it **won't be good for your grade**, even if the code is perfect.